

# A Critical Linear Bound with Logarithmic Savings for Glazer’s Time-Limited Fair-Coin Problem

Christopher D. Long

[galizur@gmail.com](mailto:galizur@gmail.com)

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## Abstract

Let a coin have unknown bias  $p \in (0, 1)$ , and let  $L$  be the length of its maximal initial constant run. We construct a deterministic, bias-independent fair-bit extractor satisfying, simultaneously,

$$T(L) \leq \lceil C_* L \rceil, \quad T(L) \leq C_* L - \log_5 L + 6, \quad C_* := 1 + \frac{2 \log \phi}{\log 5} = 1.597987435665 \dots,$$

where  $\phi = (1 + \sqrt{5})/2$ . The construction separately balances dyadic blocks at horizon  $2N$ . A parity-character folding identity supplies a real center; an elementary binomial-mixture estimate controls its positive-level norms; and a boundary-preserving integer rounding moves each interior coefficient by at most five. The argument is uniform for dyadic  $N \geq 128$ , and six smaller blocks have exact integer certificates. We prove that  $C_*$  is the optimal leading coefficient for all separately balanced horizon- $2N$  rules, without assuming complement symmetry, and that this lower bound persists with an  $o(N)$  extra horizon. An exact inequality for the deadlines bounds the possible logarithmic saving. The extractor also works for exchangeable binary sources with almost surely finite  $L$ . Finally, a parity obstruction characterizes why the block length must be a power of two at horizon  $2N$ .

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# 1 Statement of the problem and main result

Let  $X_1, X_2, \dots$  be independent coin flips with

$$\mathbb{P}(X_j = 0) = p, \quad \mathbb{P}(X_j = 1) = q := 1 - p,$$

where  $p \in (0, 1)$  is arbitrary and unknown to the rule. On every nonconstant sequence, let  $L$  denote the length of the maximal initial constant run. Thus the observed sequence starts with either  $0^L 1$  or  $1^L 0$ .

A deterministic extractor assigns one of two outputs, denoted  $H$  and  $T$ , after reading finitely many flips. It is *fair* if

$$\mathbb{P}_p(H) = \mathbb{P}_p(T) = \frac{1}{2} \quad \text{for every } p \in (0, 1).$$

Let  $T(L)$  be an upper bound for the number of flips used on sequences whose initial run has length  $L$ .

Put

$$\phi := \frac{1 + \sqrt{5}}{2}, \quad \beta := \frac{2 \log \phi}{\log 5}, \quad \alpha := 1 - \beta, \quad C_* := 1 + \beta = 2 - \alpha. \quad (1)$$

Numerically,

$$\beta = 0.597987435665\dots, \quad \alpha = 0.402012564334\dots, \quad C_* = 1.597987435665\dots$$

For each power of two  $N \geq 2$ , define the integer saving

$$s_N := \max\{0, \lfloor \log_5 N \rfloor - 3\}. \quad (2)$$

The logarithm  $\log_5 x$  means  $\log x / \log 5$ .

**Theorem 1.1** (Critical bound with logarithmic saving). *There exists a deterministic fair extractor that stops on every nonconstant input sequence. It uses two flips when  $L = 1$ . When  $N \leq L < 2N$  with  $N \geq 2$  a power of two, it satisfies*

$$T(L) \leq \min\{2N, \lceil C_* L \rceil - s_N\}. \quad (3)$$

*In particular, for every integer  $L \geq 1$ , the same extractor satisfies*

$$T(L) \leq \lceil C_* L \rceil, \quad (4)$$

$$T(L) \leq C_* L - \log_5 L + 6. \quad (5)$$

*Its two outputs have equal mass separately on every dyadic block for every bias  $p \in (0, 1)$ .*

The classical assignments  $01 \mapsto H$  and  $10 \mapsto T$  treat  $L = 1$ ; they go back to von Neumann [11]. A substantial literature studies more efficient extraction from a source of unknown bias, typically measuring expected source cost, block efficiency, or asymptotic entropy rate; see, for example, Hoeffding–Simons [5], Elias [6], Stout–Warren [7], Peres [8], Juels–Jakobsson–Shriver–Hillyer [9], and Pae–Loui [10]. The efficiency criterion here is different: it is a pathwise worst-case deadline indexed by the observed initial-run length, and is required uniformly over the unknown bias.

The particular time-limited problem considered here was posed by Elliot Glazer in an online discussion in August 2024, initially for a loaded six-sided die, together with the question of the best linear efficiency constant [1]. There are two incompatible historical conventions for the parameter

called  $n$ . In the Puzzling discussion and in Wagon’s answer, let us write  $n_{\text{SE}}$  for the number of flips up to and including the first outcome different from the initial one. Thus

$$n_{\text{SE}} = L + 1. \quad (6)$$

For the binary case, Charles Wang gave a corrected  $2n_{\text{SE}} - 2 = 2L$  construction [2]. Glazer pointed out a flaw in the initially posted version; Wang agreed and edited the construction to make the recursion greedy. That corrected construction is the valid one cited here. Wang also proposed a recursive  $7n_{\text{SE}}/4$  refinement, but that refinement was subsequently found to be invalid; Wagon’s later account records that Peter Winkler pointed out the error and that Wang acknowledged it [3]. Stan Wagon subsequently reported using the problem as a Macalester Problem of the Week; several readers found a recursive solution, and Wagon and Peter Winkler found a formulaic construction. Wagon’s posted account gives the bound  $2n_{\text{SE}} - 3 = 2L - 1$  for the binary problem [3].

Glazer later posed the coin version as Problem 12592 in the *American Mathematical Monthly* [4]. There the symbol  $n$  has a different meaning: the “critical value” is the length of the maximal initial constant run. Writing this historical parameter as  $n_{\text{AMM}}$ , we have

$$n_{\text{AMM}} = L = n_{\text{SE}} - 1.$$

Consequently the two earlier bounds become

$$2n_{\text{SE}} - 2 = 2L = 2n_{\text{AMM}}, \quad 2n_{\text{SE}} - 3 = 2L - 1 = 2n_{\text{AMM}} - 1.$$

Thus the Monthly requests  $2n_{\text{AMM}}$  and  $\max\{2, 2n_{\text{AMM}} - 1\}$  are exactly the same two bounds after translating conventions. This paper uses  $L$  throughout.

For  $L \geq 2$  we balance the events  $N \leq L < 2N$  individually. This is a sufficient construction principle, not an assumption on all possible fair extractors. Section 11 proves optimality of the leading coefficient within the separately balanced horizon- $2N$  class, and Section 12 treats sublinear additional horizons. Neither result asserts unrestricted optimality when different blocks can cancel each other’s imbalance.

The distinction between an observed stopping time and a worst-case profile is important:  $T(L)$  bounds every continuation with that initial run length. A universally fair extractor cannot stop at a constant prefix  $0^t$  or  $1^t$ . Indeed, an output assigned to  $0^t$  would have probability at least  $p^t$ , exceeding  $1/2$  for  $p$  sufficiently near one; the other case is symmetric. Thus observing the first change, at flip  $L + 1$ , is unavoidable.

## 2 The dyadic packet reduction

Fix a dyadic integer  $N \geq 2$ , put

$$n := N - 1,$$

and consider initial-run lengths  $N + i$ , where  $0 \leq i \leq n$ . Pad all decisions in this block to the common horizon  $2N$ . After the first change, the full tail then has length  $n - i$ .

For a tail  $w \in \{0, 1\}^{n-i}$ , pair

$$0^{N+i}1w \quad \text{with} \quad 1^{N+i}0\bar{w},$$

where  $\bar{w}$  is the bitwise complement. If  $w$  has  $r$  zeros, then a positively signed pair has probability difference

$$p^{N+i+r}q^{N-i-r} - q^{N+i+r}p^{N-i-r}.$$

Writing  $\ell = i + r$ , this depends only on  $\ell$ . The class  $\ell = 0$  contributes zero, and fairness is obtained by giving every class  $1 \leq \ell \leq n$  zero signed discrepancy.

Suppose that at level  $i$  the final  $a_i$  tail bits are left unread. Put

$$R_i := n - i - a_i.$$

Thus  $R_i$  tail bits are read, and the actual stopping length at this level is

$$(N + i) + 1 + R_i = 2N - a_i.$$

For each read prefix  $x \in \{0, 1\}^{R_i}$  choose a sign  $\sigma_i(x) \in \{\pm 1\}$ . The same sign is used for all  $2^{a_i}$  unread completions, while the complementary branch receives the opposite output. For  $0 \leq r \leq R_i$ , let  $e_{i,r}$  be the number of positive read prefixes containing  $r$  zeros minus the number of negative such prefixes, and write

$$E_i(x) := \sum_{r=0}^{R_i} e_{i,r} x^r.$$

**Lemma 2.1** (Packet realization). *An integer vector  $(e_{i,r})_{r=0}^{R_i}$  is realizable by deterministic signs if and only if*

$$|e_{i,r}| \leq \binom{R_i}{r}, \quad e_{i,r} \equiv \binom{R_i}{r} \pmod{2} \quad (0 \leq r \leq R_i). \quad (7)$$

The signed discrepancy contributed by level  $i$  to class  $\ell$  is

$$[x^\ell] x^i (1+x)^{a_i} E_i(x).$$

Consequently, the block is fair whenever

$$\sum_{i=0}^n x^i (1+x)^{a_i} E_i(x) \quad (8)$$

is constant.

*Proof.* There are  $Q = \binom{R_i}{r}$  read prefixes with  $r$  zeros. If  $P$  of them are positive, their net coefficient is  $2P - Q$ . This proves necessity of (7); conversely, those two conditions make  $(Q + e_{i,r})/2$  an integer in  $[0, Q]$ . Declaring, for example, the lexicographically first  $(Q + e_{i,r})/2$  prefixes positive gives a deterministic realization.

A read prefix with  $r$  zeros has  $\binom{a_i}{j}$  unread completions with  $j$  zeros. Such a completion belongs to class  $i + r + j$ . Summing over  $r$  and  $j$  gives the coefficient formula. If (8) is constant, all nonzero classes have discrepancy zero. The constant class is harmless because its paired probability difference is identically zero.  $\square$

Substitute  $x = -z$  and define

$$f_{i,r} := (-1)^{i+r} e_{i,r}.$$

The capacity and parity conditions are unchanged. It is therefore enough to find integers  $f_{i,r}$  such that

$$\sum_{i=0}^n \sum_{r=0}^{R_i} f_{i,r} z^{i+r} (1-z)^{a_i} = 1 \quad (9)$$

and

$$|f_{i,r}| \leq \binom{R_i}{r}, \quad f_{i,r} \equiv \binom{R_i}{r} \pmod{2}. \quad (10)$$

**Lemma 2.2** (Fixed-horizon type balance). *Let  $\mathcal{W} \subseteq \{0, 1\}^H$  and let  $\varepsilon : \mathcal{W} \rightarrow \{\pm 1\}$  be a signing. The identity*

$$\sum_{w \in \mathcal{W}} \varepsilon(w) p^{z(w)} (1-p)^{H-z(w)} = 0 \quad (0 < p < 1)$$

*holds if and only if*

$$\sum_{\substack{w \in \mathcal{W} \\ z(w)=k}} \varepsilon(w) = 0 \quad (0 \leq k \leq H), \quad (11)$$

*where  $z(w)$  is the number of zeros in  $w$ .*

*Proof.* Write the latter signed counts as  $c_k$ . Dividing the former expression by  $(1-p)^H$  gives  $\sum_k c_k x^k = 0$  for every  $x = p/(1-p) > 0$ . A nonzero real polynomial has only finitely many roots, so every  $c_k$  is zero. The converse follows by grouping the words by  $k$ .  $\square$

### 3 The logarithmically shifted suffix profile

We first record elementary facts about the constants. The number  $\beta$  is irrational: if  $\beta = A/B$  with positive integers  $A, B$ , then  $5^A = \phi^{2B}$ , whereas

$$\phi^{2B} = F_{2B}\phi + F_{2B-1} \notin \mathbb{Q}.$$

Here  $F_0 = 0$ ,  $F_1 = 1$ , and  $F_{j+1} = F_j + F_{j-1}$ ; the identity follows inductively from  $\phi^2 = \phi + 1$ . In particular, both  $\alpha$  and  $C_*$  are irrational.

We shall use

$$\frac{2}{5} < \alpha < \frac{13}{32}. \quad (12)$$

For the first inequality it suffices to show  $\phi^{10} < 5^3$ . Since  $\phi < 13/8$ ,

$$\phi^{10} = 55\phi + 34 < 55 \cdot \frac{13}{8} + 34 = \frac{987}{8} < 125.$$

For the second, it suffices to show  $\phi^{64} > 5^{19}$ . The recurrence gives  $F_{64} = 10610209857723$  and  $F_{63} = 6557470319842$ , while  $\phi > 809/500$  follows from  $(559/250)^2 < 5$ . Hence

$$500\phi^{64} > 809F_{64} + 500F_{63} = 11862394934818907 > 9536743164062500 = 500 \cdot 5^{19}.$$

For the rest of the construction use  $s = s_N$  from (2) and set

$$a_i := \max\{0, 2N - \lceil C_*(N+i) \rceil + s\}, \quad R_i := n - i - a_i. \quad (13)$$

Equivalently,

$$a_i = \max\{0, \lfloor \alpha(N+i) \rfloor - 2i + s\}. \quad (14)$$

In particular the suffixes are specified by integers and the sign rule does not depend on the unknown bias.

**Lemma 3.1** (Saving and profile properties). *Let  $N \geq 2$  be a power of two and let  $h = \min\{i : a_i = 0\}$ . Then:*

(i)  $s_N = 0$  for  $N < 128$ . For  $N \geq 128$ ,

$$5^{s_N} \leq \frac{N}{125}, \quad 0 \leq s_N \leq \frac{N}{64}. \quad (15)$$

(ii)  $0 \leq a_i \leq n - i$  and  $a_i - a_{i+1} \in \{0, 1, 2\}$ .

(iii) For  $0 \leq i < h$ ,

$$R_i = \lfloor \beta(N + i) \rfloor - s_N, \quad (16)$$

and  $R_0, \dots, R_h$  is nondecreasing.

(iv) The stopping length is exactly

$$2N - a_i = \min\{2N, \lceil C_*(N + i) \rceil - s_N\}. \quad (17)$$

(v) For dyadic  $N \geq 128$ ,

$$k := a_0 \leq \frac{27N}{64} - 1, \quad R_0 = n - k \geq \frac{37N}{64}. \quad (18)$$

(vi) For  $N \geq 8$ ,  $h \leq N/2$ ,  $a_{n-2} = a_{n-1} = a_n = 0$ , and  $R_i \geq 2$  for  $0 \leq i \leq n - 2$ .

*Proof.* Put  $j = \lfloor \log_5 N \rfloor$ . When  $128 \leq N < 625$ ,  $j = 3$  and  $s = 0$ , so the first inequality in (15) holds. When  $N \geq 625$ ,  $s = j - 3$  and  $5^s = 5^j/125 \leq N/125$ . The elementary induction  $64(j - 3) \leq 5^j$  for  $j \geq 4$  proves the second inequality; for  $s = 0$  it is immediate. The assertion for  $N < 128$  follows from  $N < 5^4$ .

The untruncated expression in (14) drops by either one or two at each step. Truncation at zero gives (ii)'s drop assertion. If  $a_i > 0$ , irrationality and  $\alpha + \beta = 1$  give

$$R_i = N + i - 1 - \lfloor \alpha(N + i) \rfloor - s = \lfloor \beta(N + i) \rfloor - s.$$

For  $N < 128$  this is nonnegative because  $s = 0$ . For  $N \geq 128$ ,  $\beta > 19/32$  and  $s \leq N/64$  give  $R_i > 37N/64 - 1 > 0$ . Thus  $a_i \leq n - i$  in every case, including the zero-suffix levels. The displayed  $R_i$  are nondecreasing before the transition. At the transition,

$$R_h - R_{h-1} = a_{h-1} - 1 \in \{0, 1\}.$$

This proves (ii)–(iii). Assertion (iv) is the definition of  $a_i$ .

For dyadic  $N \geq 128$ ,  $13N/32$  is integral, so

$$k = \lfloor \alpha N \rfloor + s \leq \frac{13N}{32} - 1 + \frac{N}{64} = \frac{27N}{64} - 1.$$

This proves (v).

For  $N \geq 128$ , the untruncated expression at  $i = N/2$  is less than  $39N/64 - N + N/64 < 0$ . For  $8 \leq N < 128$ , the same assertion follows with  $s = 0$ . Hence  $h \leq N/2 \leq n - 2$ . Before  $h$ , the read lengths are at least  $R_0$ ; for  $8 \leq N < 128$  we have  $R_0 = \lfloor \beta N \rfloor \geq 4$ , and for larger  $N$  use (v). After  $h$ ,  $R_i = n - i$ , proving (vi).  $\square$

## 4 A parity-character folding identity

The central construction is most transparent after the substitution

$$y := \frac{x}{1 + x}, \quad u := 1 - 2y = \frac{1 - x}{1 + x}. \quad (19)$$

A parity character on  $t$  specified read coordinates has expectation  $u^t$  under independent Bernoulli- $y$  inputs. Equivalently,

$$(1+x)^R u^t = (1-x)^t (1+x)^{R-t}. \quad (20)$$

Every coefficient of the polynomial on the right has absolute value at most the corresponding coefficient of  $(1+x)^R$ .

For a polynomial  $G(u) = \sum_{t \geq 0} p_t u^t$  with  $p_t \geq 0$ , define its fold at degree  $R$  by

$$\mathcal{F}_R G(u) := \sum_{t=0}^{R-1} p_t u^t + \left( \sum_{t \geq R} p_t \right) u^R. \quad (21)$$

Set

$$F_0(u) := \left( \frac{1+u}{2} \right)^n. \quad (22)$$

We construct polynomials  $W_i, F_{i+1}$  recursively. Suppose

$$F_i(u) = \varepsilon_i \sum_t p_{i,t} u^t, \quad \varepsilon_i \in \{\pm 1\}, \quad p_{i,t} \geq 0.$$

If  $\deg F_i \leq R_i$ , put  $W_i := F_i$  and  $W_j := 0$  for  $j > i$ . Otherwise put

$$W_i := \varepsilon_i \mathcal{F}_{R_i} \left( \sum_t p_{i,t} u^t \right), \quad F_{i+1} := \frac{F_i - W_i}{y} = \frac{2(F_i - W_i)}{1-u}. \quad (23)$$

**Lemma 4.1** (One-sign recursion). *If  $\deg F_i > R_i$ , then*

$$F_{i+1}(u) = -\varepsilon_i \sum_{s \geq R_i} \left( 2 \sum_{t > s} p_{i,t} \right) u^s. \quad (24)$$

*Thus the coefficients of  $F_{i+1}$  again have one sign, and*

$$\deg F_{i+1} = \deg F_i - 1.$$

*The recursion terminates exactly at level  $h$ , and*

$$\left( \frac{1+u}{2} \right)^n = \sum_{i=0}^h y^i W_i(u). \quad (25)$$

*Proof.* For  $G(u) = \sum_t p_t u^t$ ,

$$\begin{aligned} G(u) - \mathcal{F}_R G(u) &= \sum_{t > R} p_t (u^t - u^R) \\ &= -(1-u) \sum_{s \geq R} \left( \sum_{t > s} p_t \right) u^s. \end{aligned}$$

This proves (24). Initially  $\deg F_0 = n$ , and every nonterminal division lowers the degree by one. Hence  $\deg F_i = n - i$ . By definition,  $R_i = n - i - a_i$ , so the first level at which  $\deg F_i \leq R_i$  is exactly the first level with  $a_i = 0$ , namely  $h$ . Finally,  $F_i = W_i + y F_{i+1}$ ; iteration gives (25).  $\square$

Write

$$W_i(u) = \sum_{t=0}^{R_i} w_{i,t} u^t, \quad M_i := \sum_{t=0}^{R_i} |w_{i,t}|.$$

Define a real packet polynomial

$$E_i^{(0)}(x) := (1+x)^{R_i} W_i\left(\frac{1-x}{1+x}\right) = \sum_{t=0}^{R_i} w_{i,t} (1-x)^t (1+x)^{R_i-t}. \quad (26)$$

After replacing  $x$  by  $-z$ , write

$$E_i^{(0)}(-z) = (-1)^i \sum_{r=0}^{R_i} u_{i,r} z^r.$$

**Proposition 4.2** (Exact real center). *The coefficients  $u_{i,r}$  satisfy*

$$\sum_{i=0}^n \sum_{r=0}^{R_i} u_{i,r} z^{i+r} (1-z)^{a_i} = 1. \quad (27)$$

Moreover,

$$|u_{i,r}| \leq M_i \binom{R_i}{r}, \quad (28)$$

$u_{0,0} = 1$ , and  $u_{i,r} = 0$  for all  $i > h$ .

*Proof.* Since  $(1+u)/2 = 1-y$ , equation (25) reads

$$(1-y)^n = \sum_i y^i W_i(u).$$

Substitute  $y = x/(1+x)$  and multiply by  $(1+x)^n$ . Because  $n-i-R_i = a_i$ , the result is

$$1 = \sum_i x^i (1+x)^{a_i} E_i^{(0)}(x).$$

Replacing  $x$  by  $-z$  gives (27).

For fixed  $t$ , every coefficient of  $(1-x)^t (1+x)^{R_i-t}$  is a signed sum of exactly  $\binom{R_i}{r}$  terms of magnitude one. The triangle inequality gives (28). Finally,  $u_{0,0} = W_0(1) = F_0(1) = 1$ , and termination gives the last assertion.  $\square$

## 5 Elementary binomial estimates and logarithmic slack

All expectations in this section are finite binomial sums. The estimate below uses neither Fourier inversion nor a concentration theorem.

Let  $X \sim \text{Bin}(n, 1/2)$ . For  $i \geq 1$ , define

$$\binom{m_+}{i} := \begin{cases} \binom{m}{i}, & m \geq i, \\ 0, & m < i. \end{cases}$$

**Lemma 5.1** (Exact folded mass). *For  $1 \leq i \leq h$ ,*

$$M_i = 2^i \mathbb{E} \left( \binom{X - R_{i-1} +}{i} \right). \quad (29)$$

Also  $M_0 = 1$ .

*Proof.* Let  $p_{0,x} = 2^{-n} \binom{n}{x}$ . Repeated use of (24), together with the monotonicity of  $R_0, \dots, R_h$ , gives

$$p_{i,s} = 2^i \sum_{x>s} p_{0,x} \binom{x-s-1}{i-1} \quad (s \geq R_{i-1}). \quad (30)$$

This follows by induction: summing (30) over  $t > s$  uses

$$\sum_{t=s+1}^{x-1} \binom{x-t-1}{i-1} = \binom{x-s-1}{i}.$$

Summing (30) over  $s \geq R_{i-1}$  and applying the same hockey-stick identity gives

$$M_i = 2^i \sum_x p_{0,x} \binom{x-R_{i-1}}{i},$$

which is (29). The equality  $M_0 = 1$  follows directly from (22) and the fold preserving the sum of absolute coefficients.  $\square$

**Lemma 5.2** (An elementary binomial atom bound). *Let  $Z \sim \text{Bin}(n, \eta)$ , where  $0 < \eta < 1$ , and put  $\rho = 2 \min\{\eta, 1 - \eta\}$ . Then*

$$\max_k \mathbb{P}(Z = k)^2 \leq \frac{1}{(n+1)\rho}. \quad (31)$$

*Proof.* Write  $c_m = 2^{-m} \binom{m}{\lfloor m/2 \rfloor}$  for the largest atom of  $\text{Bin}(m, 1/2)$ . The usual consecutive-coefficient ratio locates this maximum. We have

$$c_m^2 \leq \frac{1}{m+1}. \quad (32)$$

Indeed,  $c_0 = 1$  and  $c_{2r+2} = \frac{2r+1}{2r+2} c_{2r}$ ; induction for even indices uses  $(2r+1)(2r+3) < (2r+2)^2$ . For odd indices,  $c_{2r+1} = \frac{2r+1}{2r+2} c_{2r}$ , and  $(2r+1)/(2r+2)^2 \leq 1/(2r+2)$ .

By symmetry it suffices to consider  $\eta \geq 1/2$ , so  $\rho = 2(1 - \eta)$ . Generate each trial by using a fair Bernoulli trial with probability  $\rho$  and the deterministic value one otherwise. If  $M \sim \text{Bin}(n, \rho)$  counts the fair trials, then conditionally on  $M = m$  the sum has distribution  $n - m + \text{Bin}(m, 1/2)$ . This mixture identity also follows by expanding

$$(1 - \eta + \eta z)^n = \left( (1 - \rho)z + \rho \frac{1+z}{2} \right)^n.$$

Consequently, finite Cauchy–Schwarz and (32) give

$$\max_k \mathbb{P}(Z = k)^2 \leq (\mathbb{E} c_M)^2 \leq \mathbb{E} c_M^2 \leq \mathbb{E} \frac{1}{M+1}.$$

Finally,  $\binom{n}{m}/(m+1) = \binom{n+1}{m+1}/(n+1)$  gives the exact identity

$$\mathbb{E} \frac{1}{M+1} = \frac{1 - (1 - \rho)^{n+1}}{(n+1)\rho} \leq \frac{1}{(n+1)\rho}.$$

The formula remains valid when  $\rho = 1$ . This proves the lemma.  $\square$

Put

$$p_* := 1 - \frac{1}{\sqrt{5}}, \quad q_* := \frac{1}{\sqrt{5}}, \quad \eta := \frac{\sqrt{5}}{1 + \sqrt{5}}. \quad (33)$$

We use the identities

$$\frac{2}{\sqrt{5}-1} = \frac{1+\sqrt{5}}{2} = \phi, \quad (\sqrt{5})^\beta = \phi. \quad (34)$$

**Lemma 5.3** (Tilted mass identity). *For  $i \geq 1$  define, on integers,*

$$b_i(m) := \begin{cases} \binom{m}{i} p_*^i q_*^{m-i}, & m \geq i, \\ 0, & m < i. \end{cases}$$

*If  $Z \sim \text{Bin}(n, \eta)$  and  $1 \leq i \leq h$ , then*

$$M_i = (\sqrt{5})^{\beta(n+i)-R_{i-1}} \mathbb{E} b_i(Z - R_{i-1}). \quad (35)$$

*Moreover, for every finite integer  $K \geq i$ ,*

$$\sum_{m=i}^K b_i(m) \leq \frac{1}{p_*}. \quad (36)$$

*Proof.* For  $R = R_{i-1}$  and  $x - R \geq i$ ,

$$2^i \binom{x-R}{i} = \phi^i (\sqrt{5})^{x-R} b_i(x-R).$$

Also exponential tilting of  $X \sim \text{Bin}(n, 1/2)$  gives, for every function  $g$  on its finite support,

$$\mathbb{E}(\sqrt{5})^X g(X) = \phi^n \mathbb{E} g(Z).$$

Substituting these identities into (29) and using  $(\sqrt{5})^\beta = \phi$  proves (35).

For (36), the number  $p_* b_i(m) = \binom{m}{i} p_*^{i+1} q_*^{m-i}$  is the probability that the  $(i+1)$ st success occurs at trial  $m+1$  in  $K+1$  Bernoulli trials. These events are disjoint as  $m$  ranges from  $i$  to  $K$ , so their probabilities sum to at most one. Equivalently, this is the finite identity

$$p_* \sum_{m=i}^K b_i(m) = \sum_{j=i+1}^{K+1} \binom{K+1}{j} p_*^j q_*^{K+1-j} \leq 1.$$

No infinite-series argument is required. □

**Proposition 5.4** (Uniform half-norm bound). *For every dyadic  $N \geq 128$ ,*

$$M_0 = 1, \quad M_i < \frac{1}{2} \quad (1 \leq i \leq h). \quad (37)$$

*Proof.* For  $1 \leq i \leq h$ , the preceding level has positive suffix, so

$$R_{i-1} = \lfloor \beta(n+i) \rfloor - s_N, \quad \beta(n+i) - R_{i-1} < s_N + 1.$$

By (36),

$$\mathbb{E} b_i(Z - R_{i-1}) \leq \frac{1}{p_*} \max_k \mathbb{P}(Z = k).$$

Apply Lemma 5.2 with  $\rho = 2(1 - \eta) = 2/(1 + \sqrt{5})$  and  $n + 1 = N$  to obtain

$$M_i^2 < \frac{5(1 + \sqrt{5})}{2p_*^2} \frac{5^{s_N}}{N} < 27 \frac{5^{s_N}}{N} \leq \frac{27}{125} < \frac{1}{4}. \quad (38)$$

Every numerical comparison here is elementary:  $\sqrt{5} < 9/4$  and  $p_* > 11/20$  imply

$$\frac{5(1 + \sqrt{5})}{2p_*^2} < \frac{3250}{121} < 27,$$

and  $5^{s_N} \leq N/125$  is (15). The equality  $M_0 = 1$  was proved in Lemma 5.1. □

## 6 Boundary-preserving integral parity completion

We now prove an abstract rounding lemma. For packet sites write

$$v_{i,r} := z^{i+r}(1-z)^{a_i}.$$

Put

$$d_i := a_i - a_{i+1} \in \{0, 1, 2\}.$$

For  $0 \leq i < n$  and  $1 \leq r \leq R_i$ , define the integral relation

$$K_{i,r} := \mathbf{e}_{i,r} - \sum_{j=0}^{d_i} (-1)^j \binom{d_i}{j} \mathbf{e}_{i+1,r-1+j}. \quad (39)$$

All indicated child sites exist because  $R_{i+1} = R_i + d_i - 1$ . The represented polynomial is zero.

**Lemma 6.1** (Integral kernel basis). *The relations  $K_{i,r}$  form a basis, over  $\mathbb{R}$ , of the kernel of*

$$A : (c_{i,r}) \mapsto \sum_{i,r} c_{i,r} v_{i,r} \in \mathbb{R}[z]_{\leq n}.$$

Moreover, there exists an integral vector  $q = (q_{i,r})$  such that

$$Aq = 1, \quad q_{i,r} \equiv \binom{R_i}{r} \pmod{2}. \quad (40)$$

*Proof.* The boundary columns

$$v_{i,0} = z^i(1-z)^{a_i}, \quad 0 \leq i \leq n,$$

form a unitriangular basis of  $\mathbb{R}[z]_{\leq n}$ . Thus  $A$  has rank  $n+1$ . The number of relations (39) is

$$\sum_{i=0}^{n-1} R_i = \sum_{i=0}^n (R_i + 1) - (n+1),$$

the dimension of the kernel. They are linearly independent because  $K_{i,r}$  has coefficient one at its parent site  $(i, r)$  and all remaining coordinates lie at level  $i+1$ .

For the second assertion, start with the integer vector

$$c_{i,r} := \binom{R_i}{r}.$$

Modulo two, its polynomial image is

$$\begin{aligned} Ac &\equiv \sum_{i=0}^n z^i(1+z)^{a_i+R_i} \\ &= \sum_{i=0}^n z^i(1+z)^{n-i} \\ &= (1+z)^{n+1} - z^{n+1} \\ &\equiv 1 \pmod{2}, \end{aligned}$$

because  $n+1 = N$  is a power of two. Hence  $Ac = 1 + 2H$  for some  $H \in \mathbb{Z}[z]_{\leq n}$ . The unitriangular boundary basis expresses  $H$  as an integral combination of the  $v_{i,0}$ . Subtract twice that combination from  $c$  to obtain  $q$ .  $\square$

The following lemma is the integral step that permits the critical coefficient. Its exact treatment of the two packet boundaries is essential.

**Lemma 6.2** (Five-unit lattice rounding). *Assume that  $N \geq 8$  is a power of two. Let an integer profile satisfy  $0 \leq a_i \leq n - i$ ,  $a_i - a_{i+1} \in \{0, 1, 2\}$ ,  $a_{n-2} = a_{n-1} = a_n = 0$ , and  $R_i = n - i - a_i \geq 2$  for  $i \leq n - 2$ . Let  $u = (u_{i,r})$  be a real solution of (27) such that*

$$|u_{i,r}| \leq \binom{R_i}{r}, \quad u_{0,0} = 1, \quad (41)$$

and  $u_{i,r} = 0$  on the final two levels  $i = n - 1, n$ . Then there is an integral solution  $f$  of (9) with the required packet parities such that:

(i) every lower and upper packet boundary is exact:

$$f_{i,0}, f_{i,R_i} \in \{\pm 1\}; \quad (42)$$

(ii) at every nonboundary site,

$$|f_{i,r} - u_{i,r}| \leq 5; \quad (43)$$

(iii) if  $a_{i-1} = a_i = 0$  and  $u_{i,r} = 0$ , then at every nonboundary site of level  $i$ ,

$$|f_{i,r}| \leq 2. \quad (44)$$

*Proof.* Choose  $q$  as in Lemma 6.1. Since  $A(q - u) = 0$ , there are unique real coefficients  $\gamma_{i,r}$  such that

$$q - u = \sum_{i=0}^{n-1} \sum_{r=1}^{R_i} \gamma_{i,r} K_{i,r}.$$

We shall choose even integers  $m_{i,r}$  and put

$$\delta_{i,r} := \gamma_{i,r} - m_{i,r}, \quad f := q - \sum_{i,r} m_{i,r} K_{i,r} = u + \sum_{i,r} \delta_{i,r} K_{i,r}.$$

Then  $Af = 1$ , and  $f$  has the same packet parities as  $q$ .

For every ordinary kernel coordinate, choose  $m_{i,r}$  to be a nearest even integer to  $\gamma_{i,r}$ , so that

$$|\delta_{i,r}| \leq 1. \quad (45)$$

We modify this choice along the two boundary chains, always preserving (45).

For the lower boundary, observe that

$$f_{i+1,0} = u_{i+1,0} - \delta_{i,1}.$$

Choose  $b_{i+1} \in \{\pm 1\}$  nearest to  $u_{i+1,0}$  and prescribe

$$\delta_{i,1} := u_{i+1,0} - b_{i+1}. \quad (46)$$

This has absolute value at most one. It belongs to the permitted coset  $\gamma_{i,1} + 2\mathbb{Z}$ : indeed, the lower-boundary coordinate of  $q - u$  gives

$$\gamma_{i,1} = u_{i+1,0} - q_{i+1,0},$$

and  $q_{i+1,0} - b_{i+1}$  is even. We make this prescription for  $0 \leq i \leq n-2$ ; the final overlapping variable is treated below.

For the upper boundary, write  $\eta_i := \delta_{i,R_i}$  and process  $0 \leq i \leq n-2$  in increasing order. At level zero choose  $b_0^+ \in \{\pm 1\}$  so that

$$\eta_0 = b_0^+ - u_{0,R_0}$$

has absolute value at most one. At a later level choose  $b_i^+ \in \{\pm 1\}$  so that

$$\eta_i = b_i^+ - u_{i,R_i} + (-1)^{d_{i-1}} \eta_{i-1} \quad (47)$$

has absolute value at most one. This is possible because the expression from which  $b_i^+$  is subtracted lies in  $[-2, 2]$ . It also lies in the permitted coset. Indeed, the upper-boundary coordinate of  $q - u$  is

$$q_{i,R_i} - u_{i,R_i} = \gamma_{i,R_i} - (-1)^{d_{i-1}} \gamma_{i-1,R_{i-1}},$$

so the difference between (47) and  $\gamma_{i,R_i}$  is even by induction. This makes  $f_{i,R_i} = b_i^+$ .

By the structural assumptions,  $R_{n-2} = 2$ ,  $R_{n-1} = 1$ , and  $R_n = 0$ , while  $R_i \geq 2$  for  $i \leq n-2$ . The only overlap of the lower and upper kernel variables is therefore  $(i, r) = (n-1, 1)$ . The preceding upper parameter is  $\eta_{n-2} = \delta_{n-2,R_{n-2}}$ , where  $R_{n-2} = 2$ . Because the real center vanishes on the final two levels, the upper coordinate at level  $n-1$  and the lower coordinate at level  $n$  give

$$q_{n-1,1} + q_{n,0} = -\gamma_{n-2,R_{n-2}}.$$

The left side is even. Hence  $\gamma_{n-2,R_{n-2}}$  is an even integer, and so is  $\eta_{n-2}$ ; together with  $|\eta_{n-2}| \leq 1$ , this forces

$$\eta_{n-2} = 0.$$

Moreover, the lower coordinate at level  $n$  gives  $q_{n,0} = -\gamma_{n-1,1}$ ; since  $q_{n,0}$  is odd,  $\gamma_{n-1,1}$  is an odd integer. We may therefore choose  $\delta_{n-1,1} = 1$  or  $-1$  in its allowed odd coset. The two final boundary values are then opposite signs. This completes (42) and preserves (45) everywhere.

At an arbitrary site, the error  $f - u$  consists of at most one parent term of magnitude one and the child image of the preceding level, whose total coefficient magnitude is at most

$$\sum_{j=0}^{d_{i-1}} \binom{d_{i-1}}{j} \leq 4.$$

This proves (43). If both consecutive suffix lengths are zero, then  $d_{i-1} = 0$  and the center vanishes; the error is a difference of at most two numbers of absolute value one, proving (44).  $\square$

## 7 Capacity reserve at level zero

For levels  $i \geq 1$ , Proposition 5.4 already gives proportional reserve. The level-zero digit has norm exactly one, so it requires a separate estimate.

Let

$$R := R_0 = n - k, \quad X \sim \text{Bin}(n, 1/2), \quad T := \min\{X, R\}.$$

The fold  $W_0$  gives the distribution

$$\mathbb{P}(T = t) = 2^{-n} \binom{n}{t} \quad (0 \leq t < R), \quad \mathbb{P}(T = R) = \mathbb{P}(X \geq R).$$

Let

$$K_t^{(R)}(r) := [x^r](1-x)^t(1+x)^{R-t}. \quad (48)$$

The substitution  $x = -z$  contributes the factor  $(-1)^r$  to the coefficient of degree  $r$ . Hence

$$u_{0,r} = (-1)^r \mathbb{E} K_T^{(R)}(r), \quad |u_{0,r}| = \left| \mathbb{E} K_T^{(R)}(r) \right|. \quad (49)$$

**Lemma 7.1** (Swap estimate). *For  $1 \leq t \leq R-1$  and  $1 \leq r \leq R-1$ ,*

$$\left| K_t^{(R)}(r) \right| \leq \binom{R}{r} - 2 \binom{R-2}{r-1}. \quad (50)$$

*Proof.* Fix a  $t$ -subset  $A$  of an  $R$ -element set. Then

$$K_t^{(R)}(r) = \sum_{|B|=r} (-1)^{|A \cap B|}.$$

Choose  $a \in A$  and  $b \notin A$ . On the family of  $r$ -subsets containing exactly one of  $a, b$ , the involution swapping  $a$  and  $b$  reverses the sign, so those terms cancel. There are  $2 \binom{R-2}{r-1}$  such subsets. Bounding the remaining terms by their number proves the claim.  $\square$

**Lemma 7.2** (Level-zero reserve). *For every dyadic  $N \geq 128$  and every  $1 \leq r \leq R_0 - 1$ ,*

$$\binom{R_0}{r} - |u_{0,r}| > 5. \quad (51)$$

*Proof.* Set  $R = R_0$ ,  $k = a_0$ , and  $X \sim \text{Bin}(n, 1/2)$  as above. The elementary binomial moment identities give

$$\mathbb{E} X = \frac{n}{2}, \quad \mathbb{E}(X - n/2)^2 = \frac{n}{4}.$$

They follow, for example, by summing  $x \binom{n}{x} = n \binom{n-1}{x-1}$  and  $x(x-1) \binom{n}{x} = n(n-1) \binom{n-2}{x-2}$ . For

$$\mathcal{A} = \left\{ \frac{n}{3} \leq X \leq \frac{n-1}{2} \right\},$$

symmetry for odd  $n$  gives  $\mathbb{P}(X \leq (n-1)/2) = 1/2$ . On  $X < n/3$ ,  $(X - n/2)^2 > n^2/36$ , whence

$$\mathbb{P}(X < n/3) \leq \frac{9}{n} < \frac{1}{4} \quad (n \geq 127).$$

Therefore

$$\mathbb{P}(\mathcal{A}) > \frac{1}{4}. \quad (52)$$

On this event, the truncated variable  $T = \min\{X, R\}$  equals  $X$  and lies between 1 and  $R-1$ .

For  $2 \leq r \leq R-2$ , the swap estimate gives

$$\binom{R}{r} - |u_{0,r}| \geq 2\mathbb{P}(1 \leq T \leq R-1) \binom{R-2}{r-1} > \frac{1}{2} \binom{R-2}{r-1} \geq \frac{R-2}{2} > 5,$$

because  $R \geq 37N/64$ .

For  $r = 1$ ,  $K_t^{(R)}(1) = R - 2t$  and  $M_1 = 2\mathbb{E}(X - R)_+$  imply

$$u_{0,1} = 2\mathbb{E} T - R = k - M_1 > 0.$$

Here  $k \geq 1$  follows from  $\alpha > 2/5$ , and  $M_1 < 1/2$  follows from (37). Consequently,

$$R - |u_{0,1}| = R - k + M_1 \geq n - 2k \geq \frac{5N}{32} + 1 > 5.$$

Finally,  $K_t^{(R)}(R-1) = (-1)^t(R-2t)$  yields

$$R - |u_{0,R-1}| \geq 2\mathbb{E} \min\{T, R-T\}.$$

On  $\mathcal{A}$ ,

$$\min\{T, R-T\} \geq \min\left\{\frac{n}{3}, R - \frac{n-1}{2}\right\} \geq \frac{5N}{64},$$

using (18). Together with (52), this gives  $R - |u_{0,R-1}| > 5N/128 \geq 5$ . This also proves the strict reserve at  $N = 128$ .  $\square$

## 8 All dyadic blocks above the threshold

**Theorem 8.1** (Shifted block theorem). *For every dyadic  $N \geq 128$ , there are admissible integer packets satisfying (9) for the shifted profile (13). The resulting deterministic rule on  $[N, 2N-1]$  is fair for every bias and satisfies (3).*

*Proof.* Take the real center of Proposition 4.2. At level zero its norm is one, and every positive active level has norm less than  $1/2$  by Proposition 5.4. Hence the center satisfies the packet boxes. By Lemma 3.1, all structural assumptions of Lemma 6.2 hold, and the center vanishes on the final two levels. Apply that rounding lemma.

Every packet boundary becomes  $\pm 1$  and is admissible. At level zero, Lemma 7.2 supplies more than five units of reserve at each nonboundary site. For  $1 \leq i \leq h$  and  $0 < r < R_i$ ,

$$\binom{R_i}{r} - |u_{i,r}| > \frac{1}{2} \binom{R_i}{r} \geq \frac{R_i}{2} \geq \frac{37N}{128} > 5.$$

Thus the five-unit rounding error remains inside the packet boxes. At levels  $i > h$ , the center is zero and the two consecutive suffixes are zero. The sharper rounding bound gives  $|f_{i,r}| \leq 2$  at interior sites, whose capacities are at least two. The boundary sites are already admissible. Packet realization proves fairness and (17) gives the deadline.  $\square$

## 9 The six finite blocks

The analytic proof starts at  $N = 128$ . The remaining blocks are  $N = 2, 4, 8, 16, 32, 64$ , for which  $s_N = 0$ .

**Proposition 9.1** (Exact finite certificate). *For every dyadic  $2 \leq N < 128$ , the profile (13) admits packets satisfying (9) and (10).*

*Proof.* The ancillary program `verify_glazer_critical.py` constructs these six finite witnesses. Its SHA-256 digest is

07801d62aedf91b87e122bd96f53a9bd450d64dd4e265caaff364f3da0dced61.

The packet arrays are also distributed separately in `certificates/finite_blocks.json`; their checker does not use the folding or rounding construction.

Every mathematical comparison uses integers or rational numbers. First, using rational endpoints whose squares straddle 5 and the identity

$$\log x = 2 \sum_{j \geq 0} \frac{z^{2j+1}}{2j+1}, \quad z = \frac{x-1}{x+1},$$

with the remainder bound

$$0 < 2 \sum_{j > M} \frac{z^{2j+1}}{2j+1} \leq \frac{2z^{2M+3}}{(2M+3)(1-z^2)},$$

the program certifies

$$0.5979874356654401 < \beta < 0.5979874356654403. \quad (53)$$

As an independent and purely algebraic certification of the floors, the program also checks

$$5^r < \phi^{2m} < 5^{r+1}, \quad r = \lfloor \beta m \rfloor. \quad (54)$$

These comparisons are equivalent to  $r < \beta m < r+1$  because  $\log 5 > 0$ . Using  $\phi^{2m} = F_{2m}\phi + F_{2m-1}$ , they reduce to exact integer comparisons in  $\mathbb{Q}(\sqrt{5})$ , followed by squaring only positive quantities. Finally,  $\lfloor \alpha m \rfloor = m - 1 - \lfloor \beta m \rfloor$ . The finite floor witnesses are included in the JSON certificate. Thus the decimal enclosure is not a necessary oracle for the finite floor claims. The program then uses (24) to construct the folded center with denominator  $2^{N-1}$ , performs integral kernel rounding, and checks every packet's capacity and parity. Finally, it expands the left side of (9) coefficient by coefficient over  $\mathbb{Z}[z]$  and obtains exactly  $(1, 0, \dots, 0)$ . It also checks (17) at every level. The independently checked packet arrays are therefore witnesses to the proposition.

| $N$ | $h$ | packet sites | upper bound on $\max  f - u $ | least interior margin |
|-----|-----|--------------|-------------------------------|-----------------------|
| 2   | 0   | 3            | 1                             | —                     |
| 4   | 1   | 9            | 1                             | 2                     |
| 8   | 2   | 32           | 3                             | 2                     |
| 16  | 4   | 122          | 4                             | 2                     |
| 32  | 8   | 474          | 4                             | 2                     |
| 64  | 16  | 1869         | 5                             | 2                     |

Table 1: Exact finite witnesses. The interior margin is  $\binom{R_i}{r} - |f_{i,r}|$  for  $0 < r < R_i$ ; the error bounds are integers, not rounded numerical tests.

□

## 10 Proof of the main theorem

*Proof of Theorem 1.1.* For  $L = 1$ , use  $01 \mapsto H$  and  $10 \mapsto T$ . For  $L \geq 2$ , let  $N$  be its unique dyadic scale, so  $N \leq L < 2N$ , and write  $L = N + i$ . Use Theorem 8.1 if  $N \geq 128$ , and Proposition 9.1 otherwise. Fix a definite admissible signing for each block, using the lexicographic packet realization in Lemma 2.1. Read through the first change, determine  $N$  and  $i$ , then read  $R_i$  additional bits and

output the assigned sign. The resulting rule is deterministic, uses no knowledge of  $p$ , and stops at  $2N - a_i$ .

The blocks partition the nonconstant sequences, and each block has zero signed probability for every bias. Since  $p, 1 - p < 1$ , the two infinite constant sequences have probability zero. Countable additivity now gives  $\mathbb{P}_p(H) = \mathbb{P}_p(T) = 1/2$ . The deadline is (17). Since  $s_N \geq 0$ , this implies (4).

Furthermore  $s_N \geq \log_5 N - 4$  and  $N > L/2$ , so for  $L \geq 2$ ,

$$T(L) \leq \lceil C_* L \rceil - s_N < C_* L - \log_5 L + 5 + \log_5 2 < C_* L - \log_5 L + 6.$$

The same final bound holds directly for  $L = 1$ . This proves (5).  $\square$

## 11 Optimality without complement symmetry

A rule is *separately balanced* on a block if its two outputs have equal probability on the event defining that block for every  $p \in (0, 1)$ . At horizon  $H$ , pad every decision to an  $H$ -letter word. This does not require the rule to read the padding. A per-level deadline  $t_i$  means that the sign on the level  $L = N + i$  is determined by the first  $t_i$  bits. Adaptive rules are included: take  $t_i$  to be an upper bound on all their stopping times at that level.

**Theorem 11.1** (Evaluation obstruction for arbitrary signings). *Consider a deterministic, separately balanced block  $[N, 2N - 1]$  at common horizon  $2N$ . Suppose its per-level deadlines satisfy  $N + i + 1 \leq t_i \leq 2N$ . Set*

$$a_i = 2N - t_i, \quad R_i = t_i - (N + i) - 1.$$

*No complement symmetry is assumed. For every  $0 < \theta < 1$ ,*

$$\sum_{i=0}^{N-1} \theta^i (1 - \theta)^{a_i} (1 + \theta)^{N-1-i-a_i} \geq 1. \quad (55)$$

*Proof.* Use only the branch whose initial run consists of zeros. Let  $e_{i,r}^+$  be its signed number of read tails of length  $R_i$  with  $r$  zeros, and put

$$E_i^+(x) = \sum_{r=0}^{R_i} e_{i,r}^+ x^r, \quad P_+(x) = \sum_{i=0}^{N-1} x^i (1 + x)^{a_i} E_i^+(x).$$

The coefficient of  $x^\ell$  is the signed number of zero-run words with  $N + \ell$  zeros at horizon  $2N$ . For  $\ell \geq 1$ , a word whose initial run consists of ones cannot contribute: such a word has at most  $N$  zeros. Lemma 2.2 therefore forces every nonconstant coefficient of  $P_+$  to vanish.

Its constant coefficient is the sign of the unique zero-run word with exactly  $N$  zeros, namely  $0^N 1^N$ . Hence  $P_+ \equiv \pm 1$ . Since  $|e_{i,r}^+| \leq \binom{R_i}{r}$ , evaluation at  $x = -\theta$  gives

$$1 \leq \sum_i \theta^i (1 - \theta)^{a_i} |E_i^+(-\theta)| \leq \sum_i \theta^i (1 - \theta)^{a_i} (1 + \theta)^{R_i}.$$

This is (55).  $\square$

Put

$$\theta_* := \phi^{-2} = \frac{3 - \sqrt{5}}{2}. \quad (56)$$

The identities

$$1 - \theta_* = \phi^{-1}, \quad 1 + \theta_* = \frac{\sqrt{5}}{\phi}, \quad (\sqrt{5})^\beta = \phi$$

give an exact version of the obstruction in terms of the deadlines.

**Corollary 11.2** (Exact deadline inequality). *Under the assumptions of Theorem 11.1,*

$$\boxed{\sum_{i=0}^{N-1} (\sqrt{5})^{t_i - C_*(N+i)} \geq 1 + \theta_*}. \quad (57)$$

*Proof.* For  $t_i = 2N - a_i$ , direct substitution yields

$$\theta_*^i (1 - \theta_*)^{a_i} (1 + \theta_*)^{N-1-i-a_i} = \frac{(\sqrt{5})^{t_i - C_*(N+i)}}{1 + \theta_*}.$$

Multiply (55) at  $\theta = \theta_*$  by  $1 + \theta_*$ . □

**Corollary 11.3** (Sharp leading coefficient). *Suppose fixed real constants  $C, D$  satisfy  $T(L) \leq CL + D$  for a universally fair extractor whose sufficiently large dyadic blocks are separately balanced and have horizon  $2N$ . Then  $C \geq C_*$ . Thus the infimum of affine coefficients in this class is  $C_*$ , and it is attained, for example, with  $D = 1$ .*

*Proof.* Take per-level deadlines  $t_i \leq C(N + i) + D$ . They are at least  $N + i + 1$  because no constant prefix can be terminal. If  $C < C_*$ , the left side of (57) is at most

$$N(\sqrt{5})^{(C-C_*)N+D} \longrightarrow 0,$$

contradicting its fixed positive lower bound. The construction in Theorem 1.1 supplies the matching upper bound. □

**Corollary 11.4** (Restriction on second-order savings). *Every block in Theorem 11.1 satisfies*

$$\max_{0 \leq i < N} (t_i - C_*(N + i)) \geq -2 \log_5 N + \log_{\sqrt{5}}(1 + \theta_*). \quad (58)$$

*In particular, a uniform bound*

$$T(L) \leq C_* L - \lambda \log_5 L + D$$

*within the separately balanced horizon- $2N$  class requires  $\lambda \leq 2$ . If  $\Lambda$  is the supremum of all coefficients  $\lambda$  attainable in this sense with some finite  $D$ , then*

$$1 \leq \Lambda \leq 2. \quad (59)$$

*Proof.* If the maximum in (58) is  $b$ , then (57) gives  $N(\sqrt{5})^b \geq 1 + \theta_*$ ; taking logarithms proves the first assertion. If  $\lambda > 2$ , then  $N \leq N + i$  gives

$$\sum_i (\sqrt{5})^{t_i - C_*(N+i)} \leq (\sqrt{5})^D N^{1-\lambda/2} \longrightarrow 0,$$

a contradiction. Finally, (5) attains  $\lambda = 1$ . The supremum  $\Lambda$  is not asserted to be attained. □

## 12 Sublinear additional horizons

We next allow a common horizon  $H_N = 2N + j_N$  with nonnegative integers  $j_N$ . The argument requires separate block balance, but still imposes no complement symmetry.

**Lemma 12.1** (Degree bound from separated run orientations). *Let  $N \geq 1$  and  $j \geq 0$  be integers. For a separately balanced block  $[N, 2N - 1]$  at horizon  $H = 2N + j$ , let  $t_i \in [N + i + 1, H]$  be per-level deadlines. Put  $a_i = H - t_i$ ,  $R_i = t_i - N - i - 1$ , and form the zero-run polynomial*

$$P_+(x) = \sum_{i=0}^{N-1} x^i (1+x)^{a_i} E_i^+(x)$$

as in Theorem 11.1. Then

$$\deg P_+ \leq j, \quad P_+(0) \in \{\pm 1\}.$$

*Proof.* The coefficient of  $x^\ell$  is the signed count in the zero-run branch with  $N + \ell$  zeros. A one-run word has at most  $H - N = N + j$  zeros. Thus every coefficient with  $\ell > j$  vanishes by type balance. The constant coefficient is the sign of the unique word  $0^N 1^{H-N}$ .  $\square$

**Lemma 12.2** (Elementary polynomial extrapolation). *Let  $0 < a < b$ , let  $j \geq 0$  be an integer, and let a real polynomial  $P$  have degree at most  $j$ . Then*

$$|P(0)| \leq \left( \frac{6b}{b-a} \right)^j \max_{a \leq \theta \leq b} |P(-\theta)|. \quad (60)$$

*Proof.* For  $j = 0$  the assertion is immediate. For  $j \geq 1$ , interpolate at  $-\theta_k$ , where  $\theta_k = a + k(b-a)/j$  and  $0 \leq k \leq j$ . The sum of the absolute values at zero of the Lagrange basis polynomials is at most

$$\sum_{k=0}^j \frac{b^j}{((b-a)/j)^j k!(j-k)!} = \left( \frac{2bj}{b-a} \right)^j \frac{1}{j!}.$$

The elementary inequality  $j^j/j! \leq 3^j$  completes the proof. To verify that inequality without an asymptotic factorial estimate, note that

$$\frac{(m+1)^{m+1}/(m+1)!}{m^m/m!} = \left( 1 + \frac{1}{m} \right)^m \leq \sum_{r=0}^m \frac{1}{r!} < 3,$$

using  $r! \geq 2^{r-1}$  for  $r \geq 1$ ; now induct from  $m = 1$ .  $\square$

**Theorem 12.3** (Robust leading-coefficient obstruction). *Suppose a universally fair deterministic extractor separately balances all sufficiently large dyadic blocks and has common horizons  $H_N = 2N + j_N$ , where  $j_N/N \rightarrow 0$  along the powers of two. If  $T(L) \leq CL + D$  for fixed real  $C, D$ , then  $C \geq C_*$ .*

*Proof.* Assume  $C < C_*$ . With the notation of Lemma 12.1, capacity gives, for  $0 < \theta < 1$ ,

$$|P_+(-\theta)| \leq \sum_{i=0}^{N-1} \theta^i (1-\theta)^{H_N-t_i} (1+\theta)^{t_i-N-i-1}. \quad (61)$$

Each summand is increasing as a function of  $t_i$ , because  $(1+\theta)/(1-\theta) > 1$ . Replacing  $t_i$  by its upper bound  $C(N+i) + D$  and writing  $v = i/N$  bounds the summand by

$$\frac{1}{1+\theta} \left( \frac{1+\theta}{1-\theta} \right)^D (1-\theta)^{j_N} \exp\{NG_C(\theta, v)\},$$

where

$$G_C(\theta, v) = v \log \theta + [2 - C(1 + v)] \log(1 - \theta) \\ + [C(1 + v) - 1 - v] \log(1 + \theta).$$

At the critical evaluation point,

$$G_C(\theta_*, v) = (C - C_*)(1 + v) \log \sqrt{5} < 0 \quad (0 \leq v \leq 1).$$

Because  $G_C$  is affine in  $v$ , it suffices to check its two endpoint values. By continuity, there are  $0 < a < \theta_* < b < 1$  and  $\delta > 0$  for which  $G_C(\theta, v) \leq -\delta$  whenever  $a \leq \theta \leq b$  and  $0 \leq v \leq 1$ . The remaining prefactor is bounded on  $[a, b]$ , and  $(1 - \theta)^{j_N} \leq 1$ . Thus, with a constant  $K$  independent of  $N$ ,

$$\max_{a \leq \theta \leq b} |P_+(-\theta)| \leq K N e^{-\delta N}.$$

Lemma 12.1 and Lemma 12.2 imply

$$1 = |P_+(0)| \leq \left( \frac{6b}{b-a} \right)^{j_N} K N e^{-\delta N}.$$

The logarithm of the right side tends to  $-\infty$ , since  $j_N = o(N)$  and  $\log N = o(N)$ . This is the required contradiction.  $\square$

*Remark 12.4* (Scope of the obstruction). Dropping complement symmetry does not circumvent these bounds, nor does an  $o(N)$  increase of the common block horizon. The arguments do not exclude cancellation between different dyadic blocks or separately balanced rules with larger additional horizons. In particular, they do not prove that  $C_*$  is optimal among unrestricted global extractors.

### 13 Exchangeable sources and conditional fairness

The finite combinatorial balance is stronger than fairness for independent flips. It is useful to state that consequence without any mixture representation theorem.

**Theorem 13.1** (Exchangeable-source extension). *The deterministic extractor of Theorem 1.1 is fair under every law on infinite binary sequences whose finite-dimensional distributions are invariant under coordinate permutations, provided  $L < \infty$  almost surely. Its pathwise deadlines are unchanged. Moreover, for a dyadic  $N \geq 2$ , let  $K_N$  be the number of zeros among the first  $2N$  inputs. Whenever the conditioning event has positive probability,*

$$\mathbb{P}(H \mid N \leq L < 2N, K_N = k) = \frac{1}{2}. \quad (62)$$

*The corresponding statement for  $N = 1$  follows from the pair 01, 10.*

*Proof.* By construction, each block signing has zero signed probability for all i.i.d. biases. Lemma 2.2 says that its signed count vanishes separately in every Hamming-weight class at horizon  $2N$ . Under an exchangeable finite marginal, all words in a fixed Hamming-weight class have the same probability, because a coordinate permutation maps any one of them to any other. Hence the two output events have equal probability within the intersection of the block and that class. This proves (62). Summing over the weight classes gives fairness on each block, and summing over blocks gives total fairness because  $L < \infty$  almost surely. The stopping statement is pathwise and does not depend on the source law.  $\square$

**Example 13.2.** Choose a random parameter  $P$  supported in  $(0, 1)$ , then conditionally on  $P$  generate i.i.d. inputs of bias  $P$ . The resulting source is exchangeable and has  $L < \infty$  almost surely, so the theorem applies although the inputs need not be independent after averaging over  $P$ .

*Remark 13.3.* The condition on constant sequences is necessary for almost-sure termination of this rule. Also, (62) does not assert fairness conditional on each individual value of  $L$ , nor independence of the output from the stopping time.

## 14 Why dyadic blocks occur at horizon $2N$

Here  $N$  may be any positive integer, not necessarily a power of two.

**Theorem 14.1** (Dyadic necessity and sufficiency). *The set of  $2N$ -letter words whose initial run length lies in  $[N, 2N - 1]$  admits a deterministic signing that is balanced for every bias if and only if  $N$  is a power of two.*

*Proof.* For  $0 \leq \ell < N$ , the number of zero-run words with  $N + \ell$  zeros is

$$\sum_{i=0}^{N-1} \binom{N-1-i}{\ell-i} = \binom{N}{\ell},$$

where out-of-range binomial coefficients are zero. This follows by extracting coefficients from

$$\sum_{i=0}^{N-1} x^i (1+x)^{N-1-i} = (1+x)^N - x^N.$$

For  $1 \leq \ell < N$ , these are all the words with that zero count; the one-run branch contributes only to lower counts. The reflected classes have the same cardinalities. The central class contains exactly the two words  $0^N 1^N$  and  $1^N 0^N$ .

By Lemma 2.2, a fair deterministic split exists exactly when every intermediate coefficient  $\binom{N}{\ell}$  is even. If  $N$  is a power of two, repeated squaring over  $\mathbb{F}_2$  gives  $(1+x)^N = 1+x^N$ , so all such coefficients are even. Conversely, writing  $N = \sum_j \epsilon_j 2^j$  in binary gives

$$(1+x)^N = \prod_{j:\epsilon_j=1} (1+x^{2^j}) \quad \text{in } \mathbb{F}_2[x].$$

If two or more binary digits are nonzero, the factor with the least exponent supplies a nonzero coefficient strictly between degrees zero and  $N$ . Thus  $N$  must be a power of two. When it is, split every finite Hamming-weight class into two equal parts; reading the full horizon realizes such a signing.  $\square$

**Corollary 14.2** (No fixed two-bit output in one such block). *At common horizon  $2N$ , no deterministic rule on  $[N, 2N - 1]$  can have four outputs with equal conditional probabilities within the block for every bias. In particular, the one-bit construction cannot be replaced by two uniform output bits per block without changing these requirements.*

*Proof.* Equal output probabilities for every bias force equal counts in each Hamming-weight class, by applying the polynomial argument to each pair of outputs. The central class has two words and cannot be divided into four equal integer parts.  $\square$

## Reproducibility and computational scope

The package contains this source, the compiled PDF, `verify_glazer_critical.py`, the finite packet arrays in `certificates/finite_blocks.json`, and the recorded verification output. The six arrays with  $N < 128$  are the finite witnesses used in Section 9. Larger test blocks are supplementary checks; the all- $N$  theorem is the mathematical argument of Sections 3–9, not an extrapolation from those tests.

Run the required finite checks and their independent witness checker by

```
python scripts/verify_glazer_critical.py
python scripts/verify_glazer_critical.py \
  --check-certificate scripts/certificates/finite_blocks.json
```

The recorded extended run is reproduced by

```
python scripts/verify_glazer_critical.py --extra 128 256 512 1024 \
  --export scripts/certificates/finite_blocks.json
```

Do not pass Python’s `-O` flag: assertions are part of the checker. All mathematical comparisons use exact integer or rational arithmetic; decimal displays are produced only after the corresponding exact checks. A numerical certificate is not, by itself, a proof checked by a proof-assistant kernel.

**Character expansion.** For  $c_r = [x^r](1-x)^t(1+x)^{R-t}$ , the verifier uses

$$c_0 = 1, \quad c_{-1} = 0, \quad (r+1)c_{r+1} = (R-2t)c_r - (R-r+1)c_{r-1}.$$

This follows by differentiating the product and clearing  $1-x^2$ . The script checks the recurrence independently against direct polynomial multiplication for  $0 \leq t \leq R \leq 40$ .

**Exact large-block polynomial checking.** The required finite certificates are checked coefficient by coefficient. For supplementary blocks above 128, the script may instead use the following exact test. If an integer polynomial  $Q$  has each coefficient bounded in magnitude by  $M$  and  $B > M+1$  is an integer, then

$$Q(B) = 0 \implies Q = 0.$$

Indeed, a nonzero leading integer coefficient has magnitude at least one, and its contribution dominates all lower terms, whose total magnitude is at most  $M(B^d - 1)/(B - 1) < B^d$ . For the difference from the target packet identity, the script uses the explicit bound

$$M = |\text{target}| + \sum_i 2^{a_i} \sum_r |c_{i,r}|.$$

It clears the known denominator before checking a rational center. This is an injective integer evaluation, not a probabilistic fingerprint or a congruence modulo a chosen prime.

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